More subtle but widespread racism was discovered in 2017 by the MIT professor Joy Buolamwini in commercial face-classification algorithms. She showed that these algorithms were very accurate in identifying white males, but extremely inaccurate in identifying Black females. For example, the IBM algorithm erred only 0.3 per cent of the time in identifying the gender of light-skinned males, but 34.7 per cent of the time when trying to identify the gender of dark-skinned females. As a qualitative test, Buolamwini asked the algorithms to categorise photos of the female African American activist Sojourner Truth, famous for her 1851 speech 'Ain't I a Woman?' The algorithms identified Truth as a man.[56]

When Buolamwini – who is a Ghanaian American woman – tested another facial-analysis algorithm to identify herself, the algorithm couldn't 'see' her dark-skinned face at all. In this context, 'seeing' means the ability to acknowledge the presence of a human face, a feature used by phone cameras, for example, to decide where to focus. The algorithm easily saw light-skinned faces, but not Buolamwini's. Only when Buolamwini put on a white mask did the algorithm recognise that it was observing a human face.[57]

What's going on here? One answer might be that racist and misogynist engineers have coded these algorithms to discriminate against Black women. While we cannot rule out the possibility that such things happen, it was not the answer in the case of the face-classification algorithms or of Microsoft's Tay. In fact, these algorithms picked up the racist and misogynist bias all by themselves from the data they were trained on.

To understand how this could happen, we need to explain something about the history of algorithms. Originally, algorithms could not learn much by themselves. For example, in the 1980s and 1990s chess-playing algorithms were taught almost everything they knew by their human programmers. The humans coded into the algorithm not only the basic rules of chess but also how to evaluate different positions and moves on the board. For example, humans coded a rule that sacrificing a queen in exchange for a pawn is usually a bad idea. These early algorithms managed to defeat human chess masters only because the algorithms could calculate many more moves and evaluate many more positions than a human could. But the algorithms' abilities remained limited. Since they relied on humans to tell them all the secrets of the game, if the human coders didn't know something, the algorithms they produced were also unlikely to know it.[58]